

PROFILE

- PhD-trained biostatistician with 13+ years of experience across the full clinical development lifecycle — from Phase I dose escalation to Phase III pivotal trials — available for consulting missions.
- Deep therapeutic expertise in oncology and hematology (solid tumors, CML, multiple myeloma), with hands-on experience in trial design, SAP development, advanced methodology, and regulatory interactions (FDA/EMA).
- Proven hands-on expertise in ML/AI applications to clinical data, including a co-invented patent applying machine learning to genomic data for patient prognosis prediction in oncology.

TECHNICAL EXPERTISE

- **Statistical Methodology** : Survival analysis (Cox, competing risks, frailty models), linear/logistic/mixed models, Bayesian dose-escalation (CRM, BLRM), group sequential and adaptive designs, multiple testing procedures, penalized regression (Lasso, Ridge, Elastic Net), estimand framework, sample size calculation, and event re-estimation.
- **Programming** : R (tidyverse, ggplot, RShiny, RMarkdown), Python, SAS, SQL.
- **ML/AI** : Predictive modeling, variable selection in high-dimensional spaces, internal and external model validation.
- **Medical Writing** : SAPs, CSRs, SCS/SCE, Briefing documents, health authority responses, Investigator Brochures.
- **Software & Tools** : EAST, rpact package, gsDesign packages.

WORK EXPERIENCE

Apr 2026 - Present	Independent Biostatistics Consultant, BioSPARK, Remote, France Provides independent statistical consulting across the full drug development lifecycle, from early-phase design through regulatory submission (FDA, EMA).
Feb 2023 - Mar 2026 3 years & 2 months Johnson&Johnson	Statistician in clinical trials for Multiple Myeloma, JOHNSON & JOHNSON, Remote, France > Delivered statistical design and methodology across three cutting-edge programs : > teclistamab — BCMA×CD3 bispecific, Phase II multi-arm combinations with daratumumab/lenalidomide/talquetamab. > ciltacabtagene autoleucel — BCMA-directed CAR-T, Phase II multi-arm. > ramantamig — BCMA×GPRC5D×CD3 trispecific, Phase Ib with BOIN design and Phase III design stage including event re-estimation, interim analyses for efficacy and OS harm monitoring with direct FDA/EMA interactions — covering the full design lifecycle from concept through regulatory alignment. > Authored SAPs, TFL shells, CSRs, and regulatory briefing materials; prepared written responses to FDA/EMA queries supporting regulatory submission strategy.
Nov 2019 - Jan 2023 3 years 3 months  NOVARTIS	Statistician in clinical trials for Chronic Myeloid Leukemia, NOVARTIS, Paris, FR > Delivered the full statistical package for a Phase Ib and a Phase II study on Scemblix (asciminib, first-in-class STAMP inhibitor for CML); contributed to the Phase III NDA submission package supporting FDA and EMA approval. > Led statistical methodology for a Phase I adult-to-pediatric bridging study (PIP iPSP) using a BLRM dose-escalation model. > Completed a 6-month pharmacometrics rotation contributing to PKPD and exposure-response modeling within the Scemblix development program.
Mar 2013 - Oct 2019 6 years 9 months	Statistician in clinical trials for Solid Tumors, GUSTAVE ROUSSY CANCER CENTER, Paris, FR

- › Led statistics across multiple clinical trials spanning breast (early, metastatic, TNBC, HR+), ovarian, melanoma, prostate, and pediatric oncology (Phase I–III), including Bayesian dose-escalation (TITE-CRM), precision medicine studies (SAFIR02, MAP-PYACTS), and a Phase III randomized trial in prostate cancer.
- › Developed a patented ML-based genomic signature for TNBC prognosis, owning the full analytical pipeline : gene expression extraction, data normalization (fRMA), cross-platform merging, predictive modeling, and internal/external model validation (cross-validation, bootstrap, and related techniques).
- › Published methodology research on adaptive and group sequential designs for Phase III trials in rare diseases, pediatric oncology, and biomarker-defined subtypes — covering seamless Phase II/III, platform trials, and interim analyses.
- › Generated 15+ peer-reviewed publications (Annals of Oncology, JCO, CCR, etc.) and presented at 10+ international congresses (ASCO, ESMO, AACR); invited visiting scientist at Mayo Clinic (hosted by Prof. Daniel J. Sargent).
- › Provided statistical training and capacity building for junior statisticians, clinicians, and research teams.

EDUCATION

Oct 2016 - Nov 2019
3 years 1 months



PhD in Public Health - Major Biostatistics, PARIS SACLAY UNIVERSITY, Paris, FR

- › Topic : Long-term evaluation of randomized phase III clinical trials in oncology with rare diseases and biomarker-based subtypes. Manuscript available upon request.
- › Extensive methodology work on group sequential/adaptive platform designs; taught biostatistics and public health to clinicians and undergraduates at Paris Sud University.
- › 3 Publications in highly-ranked peer-reviewed journals ([Statistics in Medicine](#), [Statistical Methods in Medical Research](#), [Computer Methods and Programs in Biomedicine](#)).

Sep 2010 - Jun 2013
2 years 10 months

Engineer in Statistics, ÉCOLE SUPÉRIEURE DE LA STATISTIQUE - UNIV. CARTHAGE, Tunis, TN
Diplôme National d'Ingénieur - Statistique et Analyse de l'Information

Sep 2012 - Oct 2013
1 years 1 months

MSc in Life science, ÉCOLE NATIONALE D'INGÉNIEURS DE TUNIS - UNIV. EL MANAR, Tunis, TN
Diplôme National de Mastère de Recherche - Traitement de l'Info. et Complexité du Vivant

LANGUAGES

Arabic Native or bilingual proficiency
French Native or bilingual proficiency
English Full professional proficiency

INTELLECTUAL PROPERTY & INNOVATION

Patent (WO2018002385A1) — hands-on ML application to medicine : built and validated a genomic signature to predict prognosis in Triple Negative Breast Cancer, owning the full pipeline from raw gene expression data to a clinically validated predictive model [[Link](#)].

SELECTED PUBLICATIONS

STATISTICAL METHODS IN MEDICAL RESEARCH, JUN. 2020 — First author  [HTTPS://PUBMED.NCBI.NLM.NIH.GOV/31354106](https://pubmed.ncbi.nlm.nih.gov/31354106)
MOHAMED AMINE BAYAR, GWÉNAËL LE TEUFF, FRANZ KOENIG, MARIE-CÉCILE LE DELEY, STEFAN MICHIELS, *Group sequential adaptive designs in series of time-to-event randomized trials in rare disease : a simulation study*

STATISTICS IN MEDICINE, AUG. 2016 — First author  [HTTPS://WWW.NCBI.NLM.NIH.GOV/PUBMED/27027783](https://www.ncbi.nlm.nih.gov/pubmed/27027783)
MOHAMED AMINE BAYAR, GWÉNAËL LE TEUFF, STEFAN MICHIELS, DANIEL J SARGENT, MARIE-CÉCILE LE DELEY, *New insights into the evaluation of randomized controlled trials for rare diseases over a long-term research horizon : a simulation study*

ANNALS OF ONCOLOGY, JAN. 2018  [HTTPS://WWW.NCBI.NLM.NIH.GOV/PUBMED/29077781](https://www.ncbi.nlm.nih.gov/pubmed/29077781)
C CRISCITIELLO, MA BAYAR, G CURIGLIANO, FW SYMMANS, C DESMEDT, H BONNEFOI, B SINN, G PRUNERI, VICIER, PIERGA, DENKERT, S LOIBL, C SOTIRIOU, S MICHIELS, F ANDRÉ, *A gene signature to predict high tumor-infiltrating lymphocytes after neoadjuvant chemotherapy and outcome in patients with triple-negative breast cancer*

Full publication list (15+ papers, 1200+ citations) : [Google Scholar profile](#).